

# Justine Sauvage

Doctor in computer science

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27 years old, French

## Education and Work Experience

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### **PhD, Centre de recherche INRIA de Paris**

*Paris*

*September 2022 - February 2026*

Games and Logic for the Verification of Cryptographic Protocols

### **Internship, CISA Helmholtz Center for Information Security**

*Saarbrücken, Germany*

*February 2022 - June 2022*

SPDM protocol modelling using Tamarin.

### **Internship, Centre de recherche INRIA de Paris**

*Paris, France*

*October 2021 - January 2022*

Squirrel prover : on the generation of axioms from cryptographic games.

### **Internship, Centre de recherche INRIA de Paris**

*Paris, France*

*March-July 2021*

Decoding algorithms for LDPC quantum code : generalisation of the small-set-flip algorithm.

### **Master degree, second year, Science, section theoretical computer science,**

*Ecole Normale Supérieure de Lyon (ENSL), Lyon, France* *September 2020-Septembre 2021*

### **Internship, LIP (Laboratoire de l'informatique du parallélisme),**

*Lyon, France*

*April-July 2020,*

Quantum cryptography: on the proof of O2H lemma.

### **Master degree, first year, Science, section theoretical computer science,**

*École Normale Supérieure de Lyon (ENSL), Lyon, France* *September 2019-April 2020*

### **Internship in Celtic Team, IRISA (Institut de Recherche en Informatique et Systèmes Aléatoires), Rennes, France**

*May-June 2019*

Blockchain and Smart Contracts in Ethereum.

Publication: Termination of Ethereum's Smart Contract in collaboration with T. Genet and T. Jensen

### **Licence (three-year university degree), Science, Section theoretical computer science,**

*Ecole Normale Supérieure de Lyon (ENSL), Lyon, France* *September 2018-May 2019*

Degree obtained on August 2019

### **Licence (first and second year), Science, Section Mathematics and Physics**

*Lycée Pierre de Fermat, Toulouse, France*

*September 2016-August 2018*

Education in mathematics, physics, chemistry and computer science at first and second year university level. Classes préparatoire aux grandes écoles.

### **Baccalauréat (French secondary school diploma/ high-school degree) Section Science,**

*Lycée Jean Pierre Vernant, Pins-Justaret, France*

*July 2016*

Degree obtained with honors.

## Languages

French: Mother tongue.

English: Level C1 (Certificate Cambridge Advanced Exam obtained on April 2019)

German: Basic

Spanish: Basic

## Teaching and popular science

Lead research workshops for teens during RJMI in 2021, 2022, and 2023.

Presentation at Alkindi, a high-school coding competition, in 2024.

Teaching: Discrete mathematics, L2, in 2022 and 2023- Tutoring exercise sessions and Python project.

## Computer Skills

- Advanced notions: Python, Ocaml, Linux, Git
- Basic notions : Coq, Isabelle/HOL, C/C++

**Others** Diving, Sailing, Guitar, and Vocal group.

## Complementary information

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## Publication

- [1] David Baelde, Adrien Koutsos, and Justine Sauvage. “Foundations for Cryptographic Reductions in CCSA Logics”. In: *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*. CCS '24. Salt Lake City, UT, USA: Association for Computing Machinery, 2024, 2814fffdfffdfffd2828. ISBN: 9798400706363. DOI: 10.1145/3658644.3690193. URL: <https://doi.org/10.1145/3658644.3690193>.
- [2] David Baelde, Adrien Koutsos, and Justine Sauvage. “Leveraging Cryptographic Simulator Synthesis for Formally Verifying the FOO E-Voting Protocol”. In: *Usenix 2026 - 35th Usenix security symposium*. Baltimore, United States, Aug. 2026. URL: <https://inria.hal.science/hal-05453231>.
- [3] Thomas Genet, Thomas Jensen, and Justine Sauvage. *Termination of Ethereum’s Smart Contracts*. Research Report. Univ Rennes, Inria, CNRS, IRISA, Apr. 2020. URL: <https://hal.inria.fr/hal-02555738>.